# LESSON 4.9
# ALGEBRAIC EXPRESSIONS WITH NEGATIVE EXPONENTS

## LESSON INTRODUCTION

**MA.8.AR.1.1** Apply the laws of exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

### LEARNING TARGETS

- I can write an equivalent expression by applying laws of exponents.

### BENCHMARK COHERENCE

**BUILDING ON**

**MA.7.NSO.1.1** Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational bases.

**WORKING TOWARD**

**MA.912.NSO.1.2** Generate equivalent algebraic expressions using the properties of exponents.

### ESSENTIAL QUESTIONS

- How do you write an equivalent expression with positive exponents for an expression that has a variable base and negative exponent?

### LESSON NARRATIVE

This lesson builds on students' prior knowledge of applying the negative exponent law to numerical expressions to add applying the law for algebraic expressions. Students continue practicing applying one or more exponent laws to generate equivalent algebraic expressions.

### COMPONENT SUMMARY

**4.9.1 WARM-UP** *(~5 MIN.)*

Generate an equivalent expression with the fewest factors possible

**4.9.2 COLLABORATION** *(15-20 MIN.)*

Rewrite algebraic expressions using positive exponents

**4.9.3 ACTIVITY** *(15 MIN.)*

Match equivalent algebraic expressions

**4.9.4 YOUR TURN!** *(5-10 MIN.)*

Generate equivalent expressions with positive exponents by applying one or more law of exponents

**4.9.5 WRAP-UP** *(~5 MIN.)*

Reflect on understanding of the learning target

# RESOURCES

### GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

### MATERIALS

- Card sort
- (Optional) Colored pencils or highlighters

### ADDITIONAL PREPARATION

- N/A

# TEACHER TIPS

### COMMON MISCONCEPTIONS

- Students may confuse the concepts of reciprocal and opposite. When applying the negative exponent law, they may take the opposite instead of the reciprocal of the expressions.

### THINGS TO CONSIDER

- This lesson serves as a review for applying the laws of exponents with algebraic expressions and provides a good opportunity to pull a small group of students who need additional support during 4.9.3 Activity or 4.9.4 Your Turn!.

### LITERACY AND LANGUAGE ROUTINES

**Take Turns**

During 4.9.3 Activity, prompt students to take turns finding each of the equivalent expression cards. Then, have the other partner explain what exponent laws would be applied to the expression. Students can take turns finding and explaining the matches. Students should discuss if they agree or disagree with their partner and explain why.

### MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

**MA.K12.MTR.5.1 Patterns and Structure**

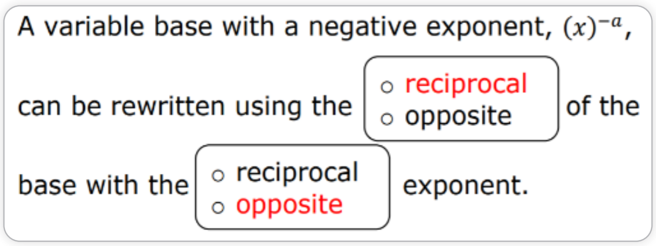
4.9.2 Collaboration elicits students to write a conjecture about rewriting an algebraic expression with a negative exponent. Through the component, students are exposed to patterns in equivalent expressions that have the same variable base and different integer exponents. Through student-led discussions, they discover how to apply the negative exponent law to a variable base with a negative exponent.

# DIFFERENTIATE INSTRUCTION

Provide auditory and visual entry points by asking students to verbally explain and show an example of reciprocal vs opposite before having them complete Questions 3c and 3f in 4.9.2 Collaboration.

4.9.4 Your Turn! provides a visual entry point by having students highlight, circle, or underline the steps they take to generate an equivalent expression.

Additional differentiation opportunities are denoted throughout the lesson guide.



# 4.9.1 WARM-UP: BELL WORK

**TEACHER MOVES**

Use student understanding to inform your instruction during this and further lessons. Make a list of misconceptions and determine which should be addressed whole group and which should be addressed in a small group with those who are struggling to be proficient.

**4.9.1 Warm-Up: Bell Work**

1. For each expression, apply the laws of exponents to generate an equivalent expression with the fewest factors possible.

   a. $\frac{k^{11}}{k^6} = k^5$

   b. $\frac{-2g^5h^8}{10gh^3} = -\frac{1}{5}g^4h^5$

   c. $z^3 \cdot z^8 = z^{11}$

   d. $\left(\frac{1}{3}x^2 \cdot y\right)^4 = \frac{1}{81}x^8y^4$

**4.9.2 Collaboration: Negative Exponents in Algebraic Expressions**

1. Rewrite each expression using positive exponents.

$6^{-4} = \frac{1}{6^4}$ $\qquad (-3)^{-2} = \frac{1}{(-3)^2}$

$\left(\frac{4}{5}\right)^{-6} = \left(\frac{5}{4}\right)^6$ $\qquad \left(\frac{-7}{8}\right)^{-1} = \frac{8}{-7}$

2. Make a conjecture on how to rewrite an algebraic expression with a negative exponent.

$x^{-3} = \frac{1}{x^3}$

3. Work with your partner to complete the following to explore negative exponents in algebraic expressions.

   a. Rewrite each expression in expanded form.

| Algebraic Expression | Expanded Form | |
|---|---|---|
| $x^3$ | $x \cdot x \cdot x$ | $\times \boxed{\frac{1}{x}}$ |
| $x^2$ | $x \cdot x$ | $\times \boxed{\frac{1}{x}}$ |
| $x^1$ | $x$ | $\times \boxed{\frac{1}{x}}$ |
| $x^0$ | 1 | |

   b. Discuss with your partner what factor could be multiplied by each expression to generate the expression in the next row. Then, fill in the blanks to the right of the table.

   c. Complete the statement.
   As the value of the exponent of $x$ ○ increases ○ decreases by 1 in the column on the left, the equivalent expression in the column on the right can be found by multiplying the previous value by the ○ opposite ○ reciprocal of $x$.

# 4.9.2 COLLABORATION: NEGATIVE EXPONENTS IN ALGEBRAIC EXPRESSIONS

## TEACHER MOVES

Monitor while students are working to determine if there are any questions that need to be debriefed before moving to the next component. This could include highlighting student voices to share rich discussions or comments overheard while they collaborated for the whole class to hear or to address common misconceptions that were evident among the majority of the class. Do not be afraid to let mistakes happen; discuss them during the transition to the next component.

## ENRICHMENT

What pattern do you notice between the exponent and the simplified algebraic expression?

## DIFFERENTIATE INSTRUCTION

Provide auditory and visual entry points by asking students to verbally explain and show an example of reciprocal vs opposite before having them answer questions 3c and 3f. This will also help ensure ELL students understand the differences in the terms so that vocabulary is not a barrier to successfully answering questions 3c and 3f.

# 4.9.2 COLLABORATION: NEGATIVE EXPONENTS IN ALGEBRAIC EXPRESSIONS (CONTINUED)

## QUESTIONING

If students are ready, ask them to write another equivalent expression for question 4d using positive exponents ($\frac{a^5}{243}$).

d. Continue the pattern.

| Algebraic Expression | Expanded Form | Equivalent Expression | |
|---|---|---|---|
| $x^0$ | $\frac{x}{x}$ | $1$ | |
| $x^{-1}$ | $\frac{1}{x}$ | $\frac{1}{x^1}$ | $\cdot \frac{1}{x}$ |
| $x^{-2}$ | $\frac{1}{x \cdot x}$ | $\frac{1}{x^2}$ | $\cdot \frac{1}{x}$ |
| $x^{-3}$ | $\frac{1}{x \cdot x \cdot x}$ | $\frac{1}{x^3}$ | $\cdot \frac{1}{x}$ |

e. Compare your conjecture in question 2 with the expression you found in Part D.
*They are the same.*

f. Complete the statement.

A variable base with a negative exponent, $(x)^{-a}$, can be rewritten using the [○ reciprocal ○ opposite] of the base with the [○ reciprocal ○ opposite] exponent.

4. For each expression, write an equivalent expression with positive exponents.

a. $x^{-4} = \frac{1}{x^4}$

b. $c^{-9} = \frac{1}{c^9}$

c. $\left(\frac{1}{f}\right)^{-2} = f^2$

d. $\left(\frac{3}{a}\right)^{-5} = \left(\frac{a}{3}\right)^5$

### 4.9.3 Activity: Equivalent Expressions Card Sort

1. Work with your partner or group to match the equivalent expressions in the set of cards. Each numbered card in the set shown below has three equivalent matches. Record the letters of the matching cards in the table.

| Numbered Cards | Equivalent Expression Cards |
|---|---|
| 1 $x^7$ | *D, K, X* |
| 2 $x^{12}$ | *F, M, AA* |
| 3 $x^{10}$ | *B, J, W* |
| 4 $x^{-9}$ | *C, L, Y* |
| 5 $\frac{1}{x^6}$ | *E, P, S* |
| 6 $\frac{1}{x^{11}}$ | *G, R, Z* |
| 7 $x^4$ | *A, N, T* |
| 8 $x^{-5}$ | *I, Q, V* |
| 9 $x^8$ | *H, O, U* |

# 4.9.3 ACTIVITY: EQUIVALENT EXPRESSIONS CARD SORT

## DIFFERENTIATE INSTRUCTION

Consider having students take turns finding each of the equivalent expression cards. Then, have the other partner explain what exponent laws would be applied to the expression. Students can take turns finding and explaining the matches. Students should discuss if they agree or disagree with their partner and explain why.

This is also an opportunity to pull a small group of students who might need additional support. The 4.9.3 Activity could be scaffolded more for those in need by the teacher leading the discussion and modeling Think Aloud strategies to identify equivalent matches.

## ENRICHMENT

How did you determine which expressions are equivalent? Which exponent laws did you use?

## 4.9.4 YOUR TURN!

**QUESTIONING**

If students finish early, challenge them to create an algebraic expression where they can apply as many laws of exponents as possible. Then, have students exchange expressions and generate an equivalent expression.

**4.9.4 Your Turn!**

1. For each expression, generate an equivalent expression with positive exponents by applying one or more laws of exponents.

   a. $\frac{9c^3}{c^{-4}} = 9c^7$

   b. $3x^{-4} \cdot 2x^{-6} = \frac{6}{x^{10}}$

2. Find the value of $m$ that makes the equation true.

   $\frac{x^4}{x^m} = x^{10}$

   $m = -6$

**4.9.5 Wrap-Up: Reflection**

1. Draw an **X** on each line to indicate how you feel about the learning target.
   *Answers will vary.*

☐ I can write an equivalent algebraic expression by applying the laws of exponents.

| I need help. I can't get started. | I'm getting there, but I need more practice. | I understand this and I feel confident. |
|---|---|---|

# 4.9.5 WRAP-UP: REFLECTION

**DIFFERENTIATE INSTRUCTION**

The qualitative data gathered from this formative assessment, coupled with what was observed during the lesson, can be used to differentiate instruction in subsequent lessons.